

MASTER'S THESIS

Title of the Thesis

at the Department of Applied Computer Science
Fulda University of Applied Sciences



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Abstract

The abstract summarizes the entire thesis in a few sentences or paragraphs. By reading it, the reader should be able to gain a general understanding of the work.

It should briefly answer the following questions:

- What is the thesis about?
- What are the open research questions addressed?
- What are the main ideas and contributions?
- What are the main results?

The results should preferably be stated qualitatively.

Use plain text where possible. Avoid formulas and references.

Although the abstract appears at the beginning of the thesis, it is usually written last.

As a general guideline, the abstract should be approximately half a page long and must not exceed one page.

Zusammenfassung

Dies ist eine kompakte Zusammenfassung der Arbeit. Sie nennt Motivation, Forschungsfrage, Methodik, wesentliche Ergebnisse und die zentrale Schlussfolgerung. Die Zusammenfassung sollte ohne Verweise auf andere Kapitel verständlich sein.

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01

Introduction

CHAPTER OVERVIEW

TODO: Add a brief chapter overview.

This chapter motivates the topic, states the research questions and objectives, describes the methodology and contributions, and outlines the structure of the thesis.

This chapter is deliberately written as continuous prose, with no section headings. Both the research questions and the contributions can be set as lists, and that is what makes them findable, so headings are not needed. As a general guideline, the introduction is three to four pages long.

Provide enough context for readers who are not yet familiar with the specific subject. Explain why the topic is relevant, which practical or scientific challenges exist, and why the problem is worth investigating.

Clearly define the problem addressed by the thesis and briefly indicate the limitations of existing solutions. This part should lead to the research questions without already describing the proposed solution in detail.

As a general guideline, this part should be approximately one page long.

The research questions follow directly from the problem stated above. They define what this thesis investigates, and every one of them must be answered by the end of it.

As a general guideline, three to six clearly formulated questions are recommended. They do not have to be phrased as questions; concise declarative statements work equally well.

Each item below is numbered RQ1, RQ2, and so on, and carries a label. Referencing that label anywhere else in the thesis prints the same name, which is how the conclusion answers each question by name.

RQ1 TODO: Replace with the first research question.

RQ2 TODO: Replace with the second research question.

RQ3 TODO: Replace with the third research question.

A research question states what is asked. An objective states what will be done in order to answer it.

Make the link explicit, so that no objective stands on its own and no research question is left without one. Keep the objectives concrete enough that it can be judged at the end whether they were achieved.

If a specific research framework or methodology is used, introduce it here and relate it to the stages of the thesis. Otherwise describe the approach in practical terms: what will be designed, built, measured, and compared.

State the criteria by which the outcome will be judged, so that the evaluation chapter has something to measure against.

A contribution is a result, not an activity. Implementing a scheduler is a task; showing that work stealing reduces tail latency by a third on up to 64 nodes is a contribution.

As a general guideline, state three to five contributions. Name for each one the chapter that delivers it, so that a reader can go straight to the evidence. Typical contributions

include a concept or design, an implementation that makes it usable, an evaluation that produces new evidence, and a publicly available artifact.

Keep this list consistent with the abstract, which states the same contributions in prose.

C1 TODO: Replace with the first contribution.

C2 TODO: Replace with the second contribution.

C3 TODO: Replace with the third contribution.

Explain how each chapter contributes to the objectives and research questions. This overview may be written as continuous text, as a structured list, or with the help of a figure.

It is often useful to indicate which chapter addresses which research question.

02

Background

CHAPTER OVERVIEW

TODO: Add a brief chapter overview.

This chapter explains the domain-specific and technical foundations needed to understand the thesis.

Explain the fundamental concepts, terminology, technologies, methods, and theoretical principles on which the thesis is based. The chapter should provide readers with the necessary context to understand the problem, the proposed approach, and the subsequent evaluation.

Possible contents include:

- definitions of relevant terms and concepts,
- theoretical foundations and established models,
- technologies, frameworks, protocols, or algorithms used in the thesis,
- the domain-specific context of the research problem,
- assumptions and technical constraints that are important for the remainder of the thesis, and
- a description of the benchmarks used.

Focus only on background information that is directly relevant to the thesis. Detailed comparisons with existing research and related approaches should normally be presented in the related work chapter.

2.1 Example Definitions

The first use of an acronym expands it automatically, so High Performance Computing (HPC) appears in full here and as a short form from the next use onwards. Declare each acronym once in `src/acronyms.tex` and never spell one out by hand.

The speedup of a parallel program on p processing elements follows from the sequential runtime $T(1)$ and the parallel runtime $T(p)$:

$$S(p) = \frac{T(1)}{T(p)} \quad (2.1)$$

Parallel efficiency relates that speedup to the number of processing elements used to reach it:

$$E(p) = \frac{S(p)}{p} \quad (2.2)$$

Equation (2.1) and Equation (2.2) are applied to measured data in Chapter 5. A program distributed across several HPC nodes with Message Passing Interface (MPI) is assessed in exactly the same way.

03

Concepts and Architecture

CHAPTER OVERVIEW

TODO: Add a brief chapter overview.

This chapter derives requirements and describes the concepts, design decisions, and resulting architecture.

This chapter presents the concrete solution approach of the thesis. It should explain the main concepts, design decisions, components, interfaces, and interactions that form the proposed system or method.

Possible contents include:

- the requirements and design goals,
- the main concepts of the proposed solution,
- the overall system architecture,
- the responsibilities of individual components,
- the interfaces and communication between components,
- relevant data flows or processing steps, and
- important architectural and design decisions.

Clearly explain how the proposed concepts address the problem introduced earlier in the thesis. Diagrams may be used to illustrate the architecture, components, interfaces, and data flows.

This chapter should focus on the specific solution developed or applied in the thesis. General theories, technologies, terminology, and other knowledge required to understand the work belong in the background chapter. Implementation-specific details should be presented in the implementation chapter.

3.1 Example Architecture Diagram

For diagrams and other freely designed graphics, draw.io / diagrams.net may be used: <https://app.diagrams.net/>

Important: Every figure generated entirely or partially using an AI-based tool must be explicitly identified and properly referenced. The use of AI must be clearly disclosed in the corresponding figure caption.

Figure 3.1 sketches a scheduler that distributes work to several compute nodes. Diagrams of this size can be drawn directly in LaTeX with TikZ, which keeps them vector-based and matches the fonts and colors of the surrounding text. The source is stored in the chapter-specific figure folder and included like any other asset:

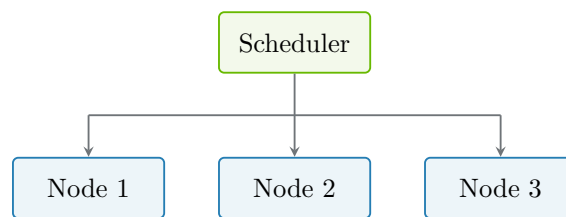


Figure 3.1: Example architecture diagram drawn with TikZ: a scheduler distributing work to three compute nodes.

04

Implementation

CHAPTER OVERVIEW

TODO: Add a brief chapter overview.

This chapter describes the realization of the concept, the technologies used, and key implementation decisions.

This chapter explains how the proposed solution was implemented in practice. It should describe the relevant technical details required to understand and reproduce the implementation.

Possible contents include:

- the selected programming languages, frameworks, libraries, and tools,
- the implementation of the individual components,
- the implementation of interfaces and communication mechanisms,
- relevant data structures, algorithms, and processing steps,
- configuration, deployment, and execution details, and
- important implementation decisions, challenges, and limitations.

Focus on details that are relevant to the thesis and necessary to understand the implementation. Avoid describing trivial code or providing a complete source-code walkthrough. Relevant code excerpts may be included where they help explain important implementation details.

The concepts and architecture chapter should describe the design of the solution, its components, responsibilities, and interactions. This chapter should focus on the concrete technical realization of that design.

4.1 Example Algorithm

Pseudocode is the right level of detail whenever the idea matters more than the syntax of a particular programming language. Algorithm 4.1 states a block-wise parallel summation in this form.

Algorithm 4.1: Block-wise parallel summation of an array.

Input: Array A with n elements, thread count p

Output: Sum of all elements of A

```

1 partial  $\leftarrow$  array of  $p$  zeros;
2 foreach thread  $t \in \{0, \dots, p-1\}$  do                                // in parallel
3   for  $i \leftarrow t \cdot n/p$  to  $(t+1) \cdot n/p - 1$  do
4   |    $partial[t] \leftarrow partial[t] + A[i];$ 
5 return  $\sum_{t=0}^{p-1} partial[t];$ 

```

05

Experiments and Evaluation

CHAPTER OVERVIEW

TODO: Add a brief chapter overview.

This chapter describes the experimental setup, metrics, and data analysis and presents the results in a reproducible manner.

This chapter reports how the solution was evaluated and what was measured. It presents results; it does not interpret them.

Possible contents include:

- the experimental setup and test environment,
- the datasets, workloads, scenarios, or test cases used,
- the selected metrics and evaluation criteria,
- the procedure used to conduct the experiments, including how often each was repeated, and
- the obtained results.

Describe the setup in enough detail that a reader could repeat the experiments and obtain the same numbers. Use tables, figures, and diagrams where they help to communicate the findings.

Keep interpretation out of this chapter. Explaining what the numbers mean, why they differ from expectations, and how they compare with other approaches is the task of the discussion chapter.

The implementation chapter describes how the solution was realized. This chapter describes how the implemented solution was examined.

5.1 Example Result Presentation

5.1.1 Tabular Results

Table 5.1 shows one possible presentation of scaling measurements. Numbers and units are set with `siunitx`, so a runtime of 120.0s and a bandwidth of 2.4 GB/s are formatted identically throughout the document.

Compute nodes	Runtime (s)	Speedup	Efficiency (%)
1	120.0	1.00	100.0
2	66.7	1.80	90.0
4	37.5	3.20	80.0
8	23.5	5.11	63.9

Table 5.1: Example scaling results for an illustrative workload.

5.1.2 Performance Plot

Figure 5.1 visualizes the same measurements and compares them with ideal scaling. Table 5.1 and Figure 5.1 thus present the same data in two forms. Unlike `\ref`, `\cref` prints the word “Table” or “Figure” itself, so a float that later changes type does not leave a wrong word behind in the running text.

Hint: Whenever suitable, use gnuplot to create plots: <https://www.gnuplot.info/>

Important: Every figure generated entirely or partially using an AI-based tool must be explicitly identified and properly referenced. The use of AI must be clearly disclosed in the corresponding figure caption.

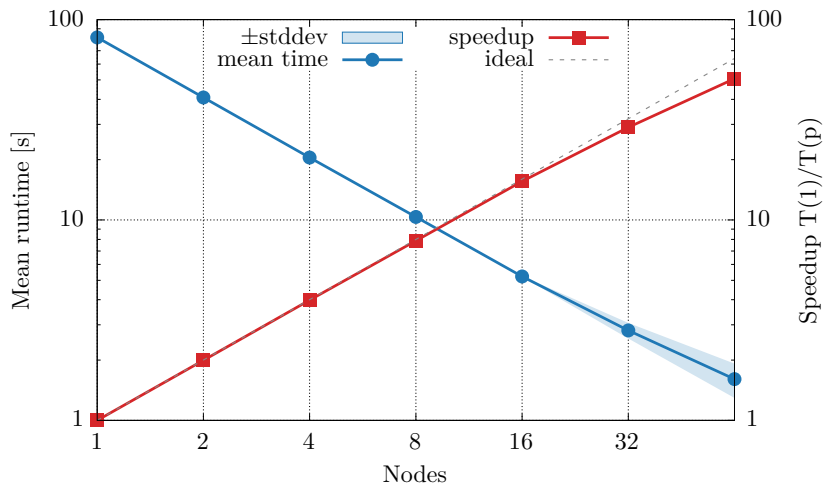


Figure 5.1: Example diagram comparing measured and ideal parallel scaling.

5.1.3 Source-Code Listing

Listing 5.1 demonstrates how source code can be included from a separate file with syntax highlighting and line numbers.

```
1  #include <stddef.h>
2  #include <omp.h>
3
4  double parallel_sum(const double *values, size_t count)
5  {
6      double sum = 0.0;
7
8      #pragma omp parallel for reduction(+:sum) schedule(static)
9      for (size_t i = 0; i < count; ++i) {
10         sum += values[i];
11     }
12
13     return sum;
14 }
```

Listing 5.1: Example OpenMP reduction for summing an array in parallel.

06

Discussion

CHAPTER OVERVIEW

TODO: Add a brief chapter overview.

This chapter interprets the results, positions them against the related work, and examines their significance, limitations, and threats to validity.

The evaluation chapter reports what was measured. This chapter explains what those measurements mean.

Possible contents include:

- the interpretation of each significant result, including the unexpected ones,
- the comparison of these results with the approaches surveyed in the related work chapter,
- an assessment of whether the objectives and requirements of the thesis were met,
- the practical implications of the findings, and
- the limitations of the proposed approach.

Explain similarities, differences, advantages, and the possible reasons for differing results. Do the results confirm, extend, or contradict existing findings?

An unexpected or negative result is a legitimate outcome and is worth discussing carefully. Explaining why something did not work is often more informative than reporting that something did.

Do not introduce new experimental results in this chapter. New results belong in the evaluation chapter, where they are presented before being interpreted here.

6.1 Threats to Validity

A thesis that names its own weaknesses is more convincing than one that claims to have none. An unacknowledged threat that an examiner spots is far more damaging than one that is stated openly.

Consider at least the following categories:

- Internal validity: could the observed effect have a cause other than the one claimed?
- External validity: how far do the results generalize beyond the tested system, workloads, and scale?
- Construct validity: do the chosen metrics actually measure the property of interest?

Typical threats in systems and performance work include measurement noise with too few repetitions, results obtained on a single machine or cluster, a workload selection that happens to favor the proposed approach, and extrapolation beyond the scale that was actually measured.

State for each threat how it was mitigated, or why it could not be.

07

Related Work

CHAPTER OVERVIEW

TODO: Add a brief chapter overview.

This chapter surveys the existing scientific work on the problem, groups it into lines of research, and identifies the gap addressed by this thesis.

Remember that this is a scientific thesis. A thorough literature review is therefore expected. Identify relevant publications, established approaches, comparable systems, and previous research concerning the problem addressed by the thesis.

The chapter should address questions such as:

- Which relevant approaches and findings already exist?
- How can they be grouped into categories or lines of research?
- What are the strengths and weaknesses of each approach?
- Which research gap or open problem remains?
- How does this thesis address that gap?

Do not merely list or summarize individual publications one after another. Compare and categorize them instead, and make the comparison explicit, for example in a table with one row per approach and one column per criterion. All statements concerning previous research must be supported by appropriate scientific references.

Suitable tools for conducting the literature search include:

- Google Scholar: <https://scholar.google.com>

An example citation of a peer-reviewed paper is [1]. An official website is cited in the same way [2].

08

Conclusion and Future Work

CHAPTER OVERVIEW

TODO: Add a brief chapter overview.

This chapter summarizes the main contributions and results, answers the research questions, and identifies directions for future work.

8.1 Conclusion

Give a concise final assessment of the work, without repeating the complete discussion and without introducing new results. State whether the objectives set out in the introduction were achieved.

Answer every research question within this text, in the order in which they were posed. Refer to each one by its number, so that a reader can match the answers to the questions in the introduction.

This is what an examiner reads to confirm that the thesis delivered what it promised. A research question that is quietly left unanswered is more visible here than anywhere else in the thesis. If a question could not be answered, say so and explain why.

8.2 Future Work

Base them on the limitations and findings of this thesis rather than listing generic ideas. Possible topics include improvements to the proposed approach, extensions of the implementation, and additional experiments or evaluations.

The threats to validity named in the discussion chapter are often the most honest source of future work.

References

- [1] J.-P. Lozi, B. Lepers, J. R. Funston, F. Gaud, V. Quéma, and A. Fedorova. “The Linux Scheduler: A Decade of Wasted Cores”. In: *Proceedings of the Eleventh European Conference on Computer Systems*. EuroSys ’16. Association for Computing Machinery, 2016, 1:1–1:16. DOI: 10.1145/2901318.2901326.
- [2] SchedMD. *Slurm Workload Manager: Overview*. URL: <https://slurm.schedmd.com/overview.html> (visited on 07/30/2026).

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List of Acronyms

HPC High Performance Computing

MPI Message Passing Interface

A

Appendix

CHAPTER OVERVIEW

TODO: Add a brief chapter overview.
This appendix collects supplementary material that supports the thesis but would interrupt the flow of the main chapters.

Declaration of Authorship

Name: First_name Last_name
Student ID: 123456
Study program: Name of the study program
Module: Master's Thesis

I hereby declare that I have written the examination work or thesis in the module named above – in the case of group work, the correspondingly marked part of the work – independently and without using any sources or aids other than those indicated.

Texts, lines of thought, concepts, graphics, and similar material taken directly or indirectly from external sources have been clearly identified as such in my work and provided with complete references to the respective source.

All quotations and all passages that were written or modified through the use of AI-based tools have been checked for correctness and marked accordingly. Documentation of the AI tools used, together with the purpose for which each was used, is attached to this work. I bear responsibility for the quality and for all contents of this work, and my own creative contribution predominates. I further declare that I have used exclusively AI-based tools whose use has been explicitly designated as a permitted aid by the examiner.

This work has not been submitted in the same or a similar form in any other examination.

Place, date

Signature